I/E/TZ

Reg. No: 23BCB0517

Final Assessment Test – April 2026

Vellore Institute of Technology

Course: BCSE305L - Embedded Systems

Class NBR(s): 1645/1646/1650/1653/1654/1657/1661/ 1665/1669/1672/1674/1678/1683/1686/1687/1689/ 1695/5204 Slot: G2+TG2

Time: Three Hours Max. Marks: 100

- KEEPING MOBILE PHONE/ANY ELECTRONIC GADGETS, EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE
- DON'T WRITE ANYTHING ON THE QUESTION PAPER

COs	CO Statements
CO1	Identify the challenges in designing an embedded system using various microcontrollers and interfaces.
CO2	To summaries the functionality of any special purpose computing system, and to propose smart solutions to engineering challenges at the prototype level.
CO3	To examine the working principle and interface of typical embedded system components, create programme models, apply various optimization approaches including simulation environment and demonstration using debugging tools.
CO4	To evaluate the working principle of serial communication protocols and their proper use, as well as to analyze the benefits and drawbacks of real-time scheduling algorithms and to recommend acceptable solutions for specific challenges.

ooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create)

Answer ALL Questions
(10 X 10 = 100 Marks)

1. Describe the PIC microcontroller architecture with diagram, focusing on ADC, CCP, PWM, WDT and interrupts. CO1 BL2

2. a) Sketch the Internal memory organization in 8051 and brief the use of SFRs namely TCON, SCON, PCON and SBUF. [5] CO1 BL2
   b) Compare and contrast the salient features of PIC and ARM. [5]

3. a) Draw an ADC Interface with 8051MC to read from 3 different sensors and also show how to select input. [5] CO1 BL3
   b) What is the role of decoders in embedded system design? Draw the diagram for interfacing a 3-bit decoder with MC8051. [5]

4. a) Illustrate the role of watchdog timer in embedded system applications. How it is implemented in 8051? [5] CO1 BL2
   b) Write an 8051 C program to transfer the message "VIT-VELLORE" serially at 9600 baud rate, 8 bit data and one stop bit, continuously. Assume Crystal frequency of 12 MHz. [5] CO1 BL3

5. In a home safety system using 8051MC, a door sensor is connected to the P1.0 pin, window sensor is connected to the P1.1 pin, and the siren is connected to P2.0. Write an 8051 C program to monitor the door and window sensor. When any one gets open, sound the siren by sending a square wave of a 100 Hz for 100ms. Illustrate with a suitable diagram and 8051 program. CO2 BL3

6. In an ATM system, a PIR door sensor (S) is connected to pin P2.7 of MC8051. If S=0, the dc motor moves CLK-wise and door closed. If S=1, the dc motor moves CCLK-wise and door opened. Draw the schematic diagram and write 8051 program. CO2 BL3

7. Implement an asynchronous finite state machine to do the data transfer between a 'Talker' and a 'Listener' connected by a data bus and 2 control signals. The Talker uses the control signal ***RTS*** (Request to Send) to indicate that valid data is available on the bus. The Listener uses the signal *ACK* (Acknowledgement) to indicate that it has accepted previous data and is ready for more. The Talker receives an internal signal '***send***' from the data generator to indicate that data is waiting to be sent. Similarly, the listener receives an internal '***recv***' signal from the consumer of data to start receiving data from the bus. The following protocol is used: CO3 BL4

Initially, both Talker and Listener are in their idle states with *RTS, ACK, send and rcv* FALSE. When the internal 'send' signal becomes TRUE. The talker places data on the bus and asserts RTS to be TRUE. A TRUE signal on RTS makes the internal 'send' signal FALSE. The talker now waits for ACK to become TRUE. When the listener sees RTS become TRUE, it checks if the '*recv*' signal is TRUE. When '*recv*' is TRUE, it latches the data and asserts ACK as TRUE. It now waits for RTS to become FALSE. When the Talker sees ACK as TRUE, it makes RTS FALSE. The listener on seeing RTS become FALSE, makes ACK FALSE. Now we are at the initial condition and transfer of next word on the bus can follow the same procedure. Show state diagrams for the talker and listener to implement this protocol. Assume that state transitions are event driven and the Talker state machine is invoked whenever there is an event on '*send or ACK*', while the listener state machine is invoked whenever there is an event on *recv or on RTS*. Also show the state list, tests and actions for both state machines. There has to be a unique test in each state which returns a binary answer.

8. a) Discuss the multi-master I2C bus communication and how to avoid bus contention. [5] CO4 BL2

b) Draw and explain bus communication used in automotive embedded system. [5]

9.a) An embedded system for a packaging plant in an industry has to be designed which is controlled by a centralized embedded system using a MC. The requirements of the packaging plant system are listed below: CO4 BL4

i. An optimal temperature (25°C) needs to be maintained inside the packaging plant.
ii. The temperature monitoring requires use of 5 different temperature sensors placed at different locations in the packaging room and then averaged for every 30 secs.
iii. At each corner of the packaging room 4 fans are provided to control the temperature. The speed of the fan is dependent upon the room temperature.
iv. Display the current temperature using LCD display connected serially.

1. Show the system design with block diagram and pseudo code for this task.
2. Illustrate the operation of the system by using DFG & CDFG.

OR

9.b) Design an embedded system using a MC to regulate glucose levels in the body of a patient with diabetes by continuously measuring the level of glucose and dispensing doses of insulin based on those measurements. The chemical glucose sensor generates 4-bit digital signal. The insulin infusion pump is controlled by a PWM signal. The system has ON/OFF switch, RESET switch, alarm speaker and LCD for blood glucose monitoring. CO4 BL4

i. Show the system block diagram including each unit in your design and pseudo code.
ii. Illustrate the operation of the system by using CDFG. Also give the use of ICE based debugging.

10.a) A RTOS scheduler maintains two ready queues, one for high priority processes and the other for low priority ones. If the high priority queue has any entries, processes are scheduled from this queue. Only if the high priority queue is empty, processes are scheduled from the low priority queue. For scheduling from either queue, least slack time first priority is used. Scheduling takes places at the beginning of each time slot of duration *T*. Ready to run processes are inserted in the appropriate ready queue before scheduling decisions are taken, so they can run in that slot itself, if selected. The following table gives the arrival time, total run time and deadlines for a few processes in units of *T*. P0 and P1 are high priority, while the rest are low priority processes. CO4 BL4

Priority	High			Low			
Process	P0	P1		P2	P3	P4	
Ready to run at	4	1		0	3	6	
Total Run Time	3	5		3	2	1	
Deadline	11	15		16	12	10	

Show the detailed scheduling of these processes at each time slot (beginning at 0), justifying why a process is chosen to run at that time, till all processes are over.

OR

10.b) i. Find an optimal value "*x*" to make it schedulable under RMA. Draw a task timeline graph for at least 3 cycles. Comment on priority inversion in this RMA scheduling and give a solution.

Task	Execution Time	Period
T1	5	15
T2	7	10
T3	*x*	20

ii. List out salient features of eCOS as a RealTime OS.

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